

RIVERINE FLOODS AND URBAN FLOOD RESILIENCE

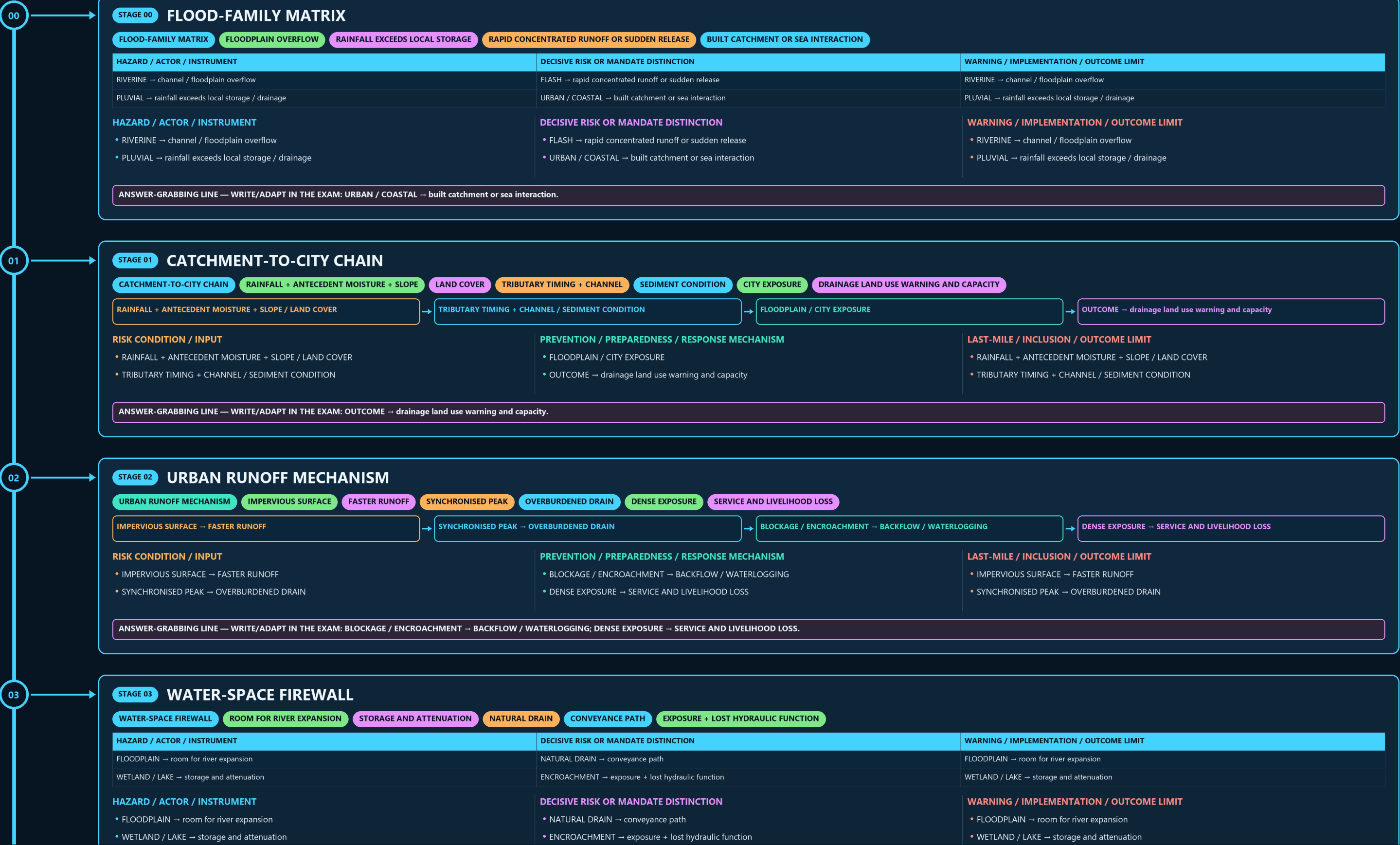
FLOOD-FAMILY MATRIX → CATCHMENT-TO-CITY CHAIN → URBAN RUNOFF MECHANISM → WATER-SPACE FIREWALL → COMPOUND COASTAL FLOOD MAP → CWC FORECAST RAIL

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g3 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BLOCKAGE / ENCROACHMENT → BACKFLOW / WATERLOGGING; DENSE EXPOSURE → SERVICE AND LIVELIHOOD LOSS.

03

STAGE 03 WATER-SPACE FIREWALL

WATER-SPACE FIREWALL ROOM FOR RIVER EXPANSION STORAGE AND ATTENUATION NATURAL DRAIN CONVEYANCE PATH EXPOSURE + LOST HYDRAULIC FUNCTION

HAZARD / ACTOR / INSTRUMENT

FLOODPLAIN → room for river expansion
WETLAND / LAKE → storage and attenuation

DECISIVE RISK OR MANDATE DISTINCTION

NATURAL DRAIN → conveyance path
ENCROACHMENT → exposure + lost hydraulic function

WARNING / IMPLEMENTATION / OUTCOME LIMIT

FLOODPLAIN → room for river expansion
WETLAND / LAKE → storage and attenuation

HAZARD / ACTOR / INSTRUMENT

- FLOODPLAIN → room for river expansion
- WETLAND / LAKE → storage and attenuation

DECISIVE RISK OR MANDATE DISTINCTION

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ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NATURAL DRAIN → conveyance path; ENCROACHMENT → exposure + lost hydraulic function.

04

STAGE 04 COMPOUND COASTAL FLOOD MAP

COMPOUND COASTAL FLOOD MAP RIVER FLOW PLUVIAL RUNOFF URBAN DRAIN DOWNSTREAM IMPEDANCE COMPOUND INUNDATION

RIVER FLOW → ESTUARY

PLUVIAL RUNOFF → URBAN DRAIN

SURGE / TIDE / WAVE → DOWNSTREAM IMPEDANCE

COMBINATION → COMPOUND INUNDATION

HAZARD / RISK / RESILIENCE CONCEPT

- RIVER FLOW → ESTUARY
- PLUVIAL RUNOFF → URBAN DRAIN

INSTITUTIONAL OR COMMUNITY RESPONSE

- SURGE / TIDE / WAVE → DOWNSTREAM IMPEDANCE
- COMBINATION → COMPOUND INUNDATION

CAPACITY / STATUS / EVIDENCE LIMIT

- RIVER FLOW → ESTUARY
- PLUVIAL RUNOFF → URBAN DRAIN

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SURGE / TIDE / WAVE → DOWNSTREAM IMPEDANCE.

05

STAGE 05 CWC FORECAST RAIL

CWC FORECAST RAIL 1 OBSERVE LEVEL 2 ANALYSE AND FORECAST 3 ISSUE SITE-SPECIFIC CATEGORY 4 ADMINISTRATION PROJECT AUTHORITY ACTS

1 OBSERVE LEVEL / DISCHARGE / INFLOW

2 ANALYSE AND FORECAST

3 ISSUE SITE-SPECIFIC CATEGORY / MESSAGE

4 ADMINISTRATION / PROJECT AUTHORITY ACTS

HAZARD / RISK / RESILIENCE CONCEPT

- 1 OBSERVE LEVEL / DISCHARGE / INFLOW
- 2 ANALYSE AND FORECAST

INSTITUTIONAL OR COMMUNITY RESPONSE

- 3 ISSUE SITE-SPECIFIC CATEGORY / MESSAGE
- 4 ADMINISTRATION / PROJECT AUTHORITY ACTS

CAPACITY / STATUS / EVIDENCE LIMIT

- 1 OBSERVE LEVEL / DISCHARGE / INFLOW
- 2 ANALYSE AND FORECAST

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: 3 ISSUE SITE-SPECIFIC CATEGORY / MESSAGE; 4 ADMINISTRATION / PROJECT AUTHORITY ACTS.

06

STAGE 06 RESERVOIR GOVERNANCE BOUNDARY

RESERVOIR GOVERNANCE BOUNDARY INFLOW FORECAST + CURRENT STORAGE OPERATING RULE DAM CONDITION DOWNSTREAM CAPACITY GATE DECISION TIMELY DOWNSTREAM WARNING DAM IS NEITHER GUARANTEE NOR SINGLE CAUSE

INFLOW FORECAST + CURRENT STORAGE

OPERATING RULE / DAM CONDITION / DOWNSTREAM CAPACITY

GATE DECISION → TIMELY DOWNSTREAM WARNING

TRAP → DAM IS NEITHER GUARANTEE NOR SINGLE CAUSE

HAZARD / ACTOR / INSTRUMENT

- INFLOW FORECAST + CURRENT STORAGE
- OPERATING RULE / DAM CONDITION / DOWNSTREAM CAPACITY

DECISIVE RISK OR MANDATE DISTINCTION

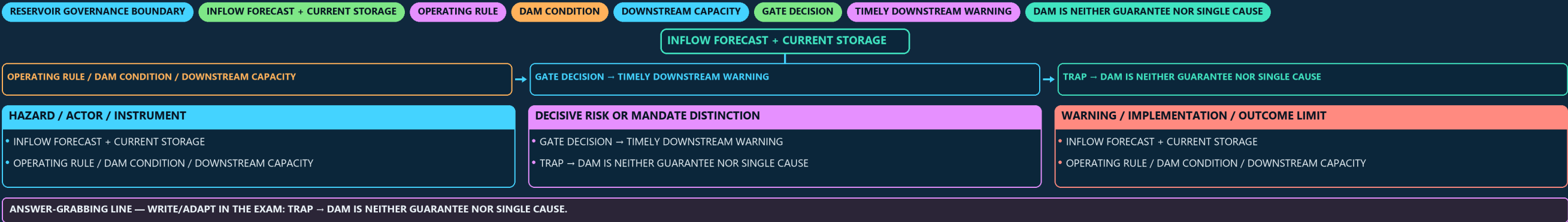
- GATE DECISION → TIMELY DOWNSTREAM WARNING
- TRAP → DAM IS NEITHER GUARANTEE NOR SINGLE CAUSE

WARNING / IMPLEMENTATION / OUTCOME LIMIT

- INFLOW FORECAST + CURRENT STORAGE
- OPERATING RULE / DAM CONDITION / DOWNSTREAM CAPACITY

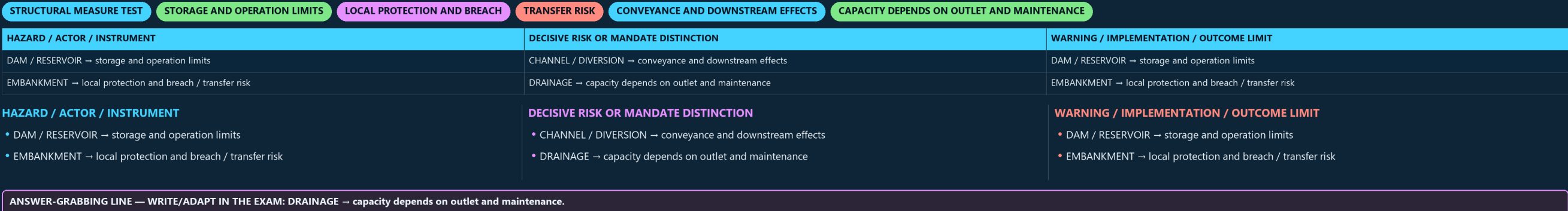
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → DAM IS NEITHER GUARANTEE NOR SINGLE CAUSE.

RESERVOIR GOVERNANCE BOUNDARY



07

STAGE 07 STRUCTURAL MEASURE TEST



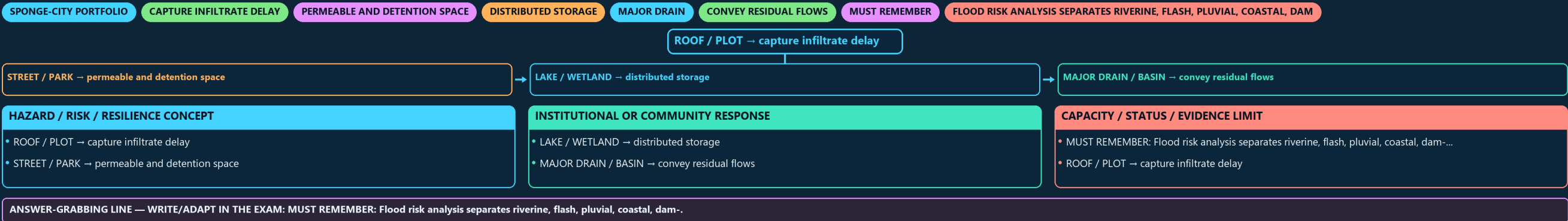
08

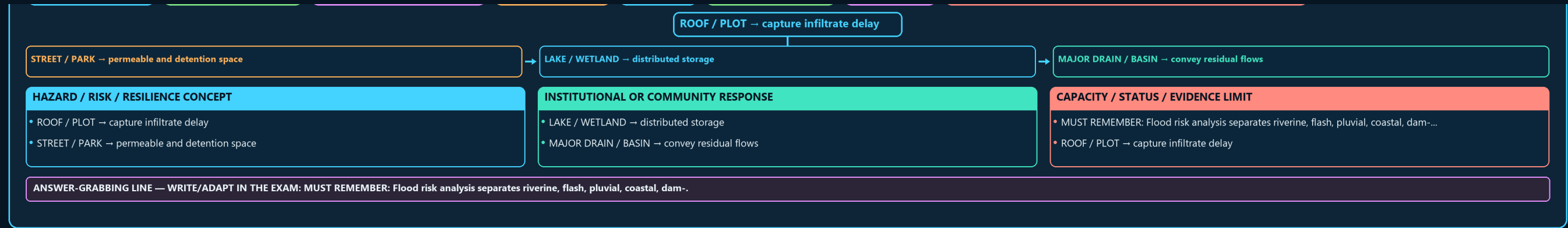
STAGE 08 URBAN PLANNING PORTFOLIO



09

STAGE 09 SPONGE-CITY PORTFOLIO





10



11



E

